

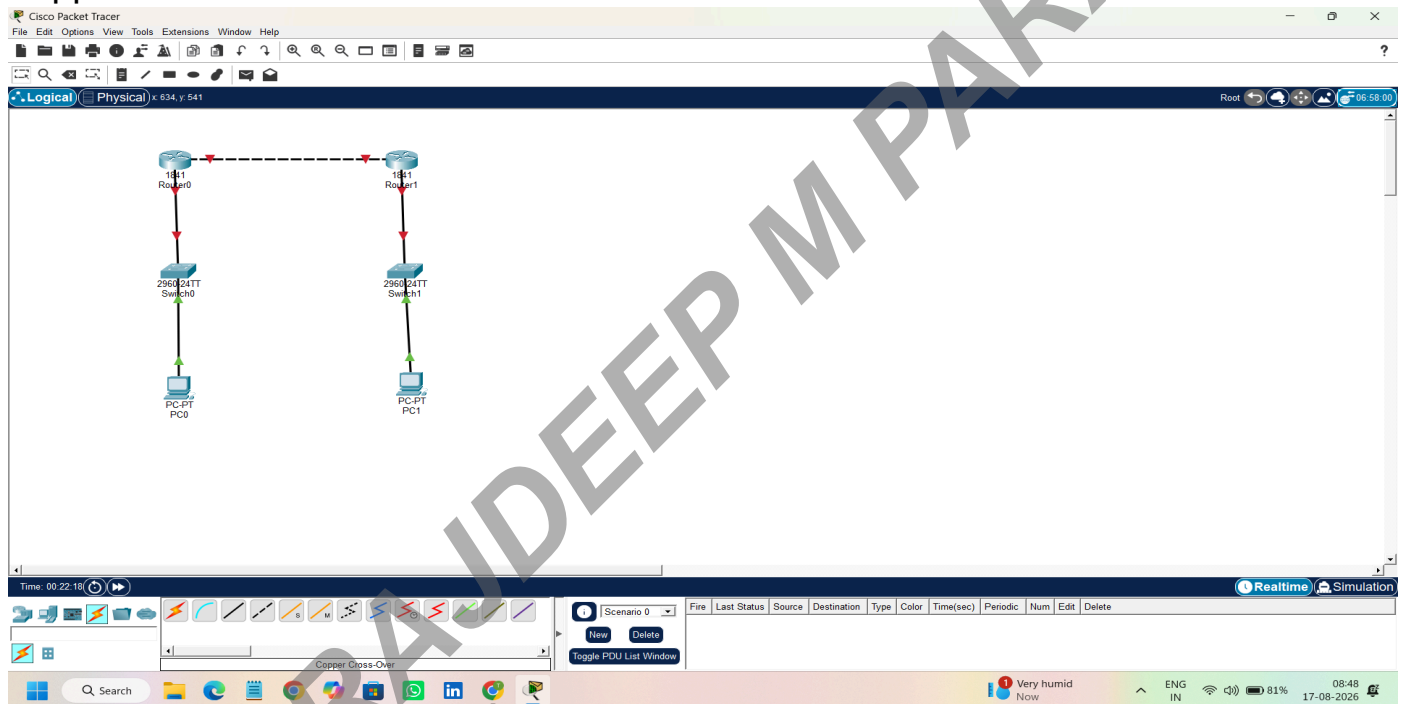
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SUBJECT NAME: T101 Cyber & Information Security Practical PRACTICAL - 6

Aim: Configure IPsec on network devices to provide secure communication and protect against unauthorised access and attacks.

Steps:-

Step 1:- Open Cisco Packet And Add 2 Routers, 2 switches, and 2 PCs And Connect Them To Copper Straight Through And Router Connect With Copper Cross Over

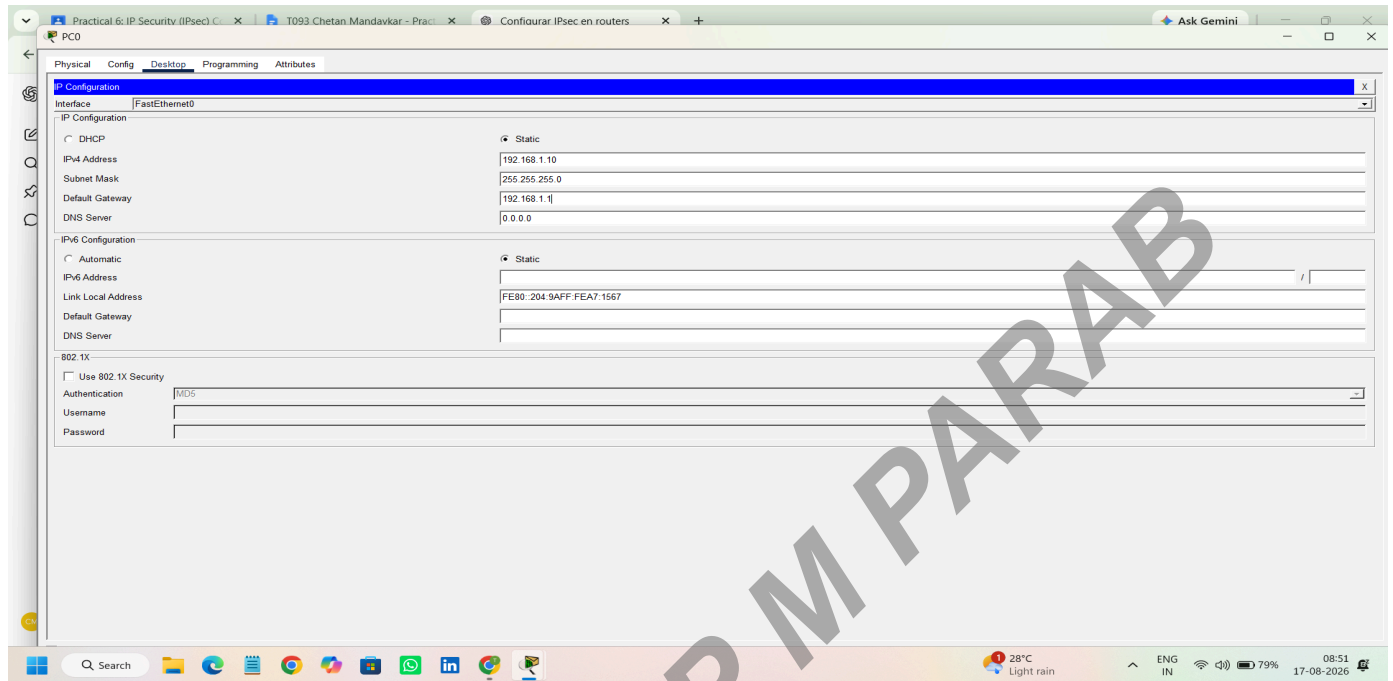


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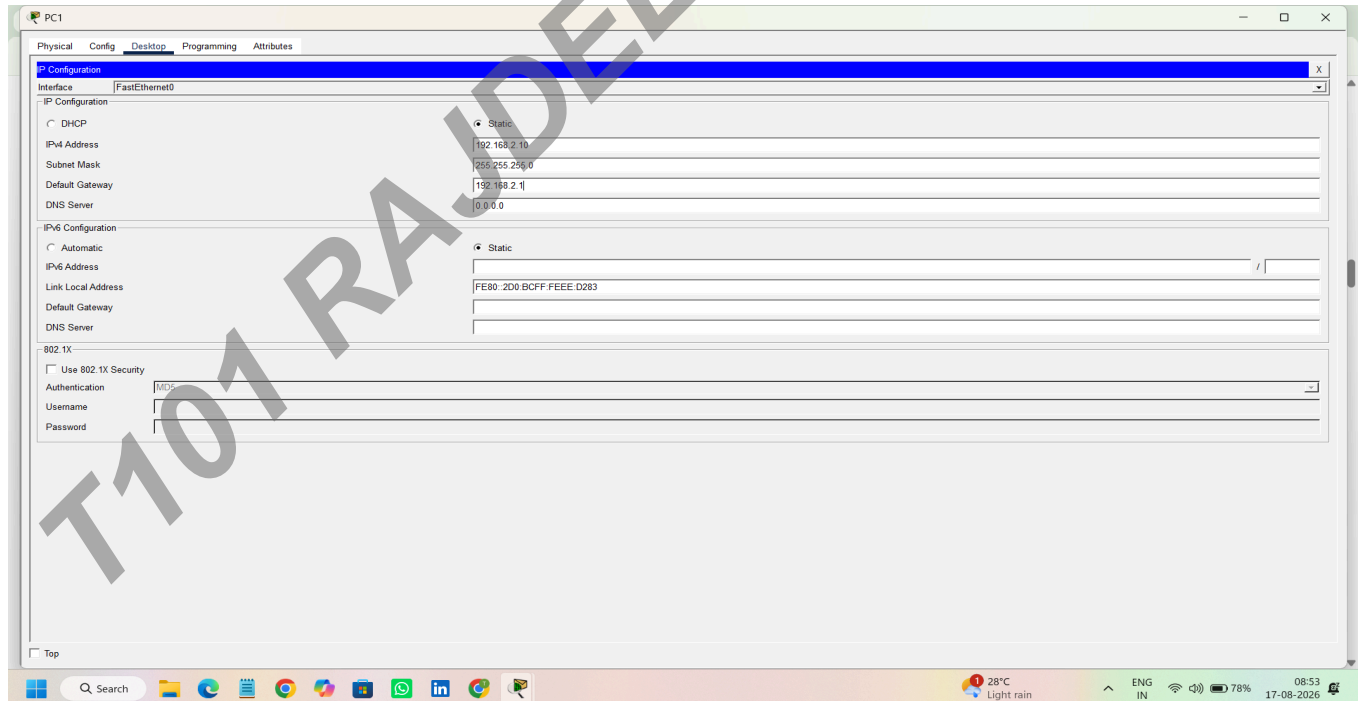
SUBJECT NAME: T101 Cyber & Information Security Practical

Step 2:- Configure PCs

PC0:-



PC1:-

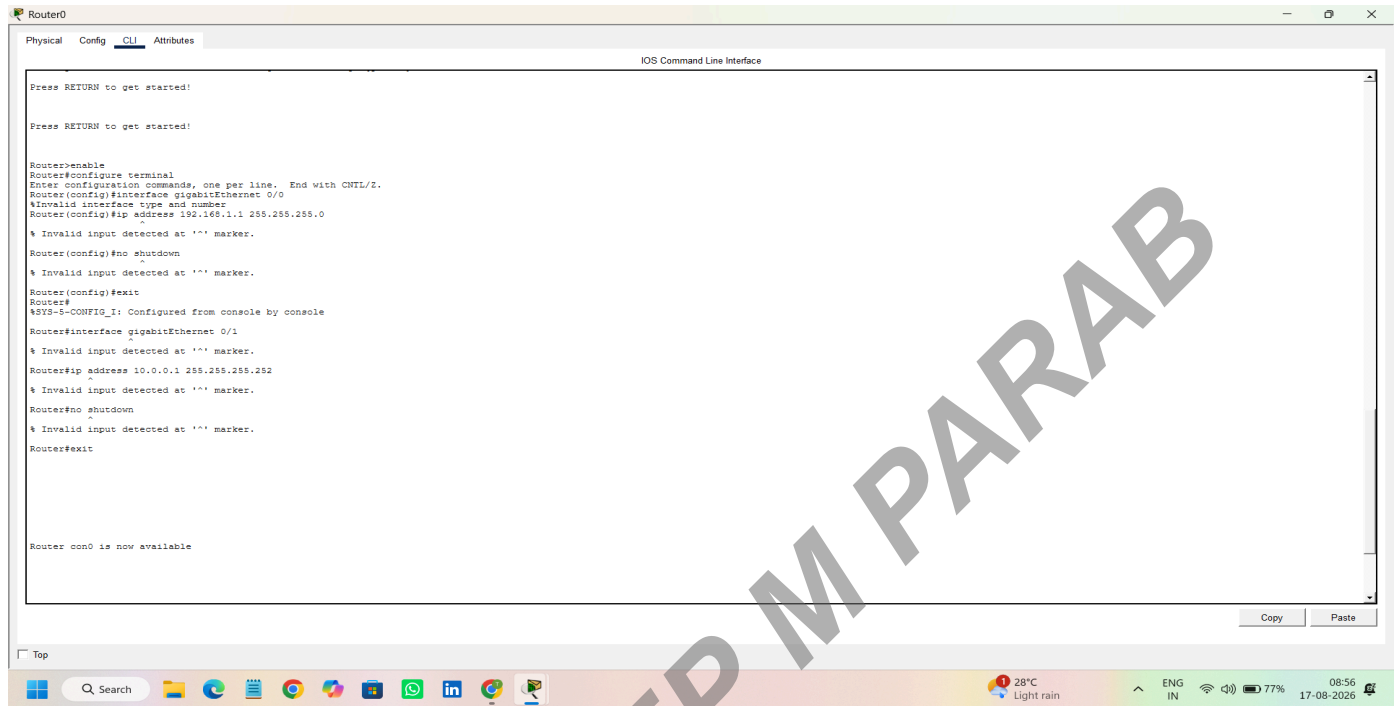


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Step 3:- Configure Routers

Router 0:-



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

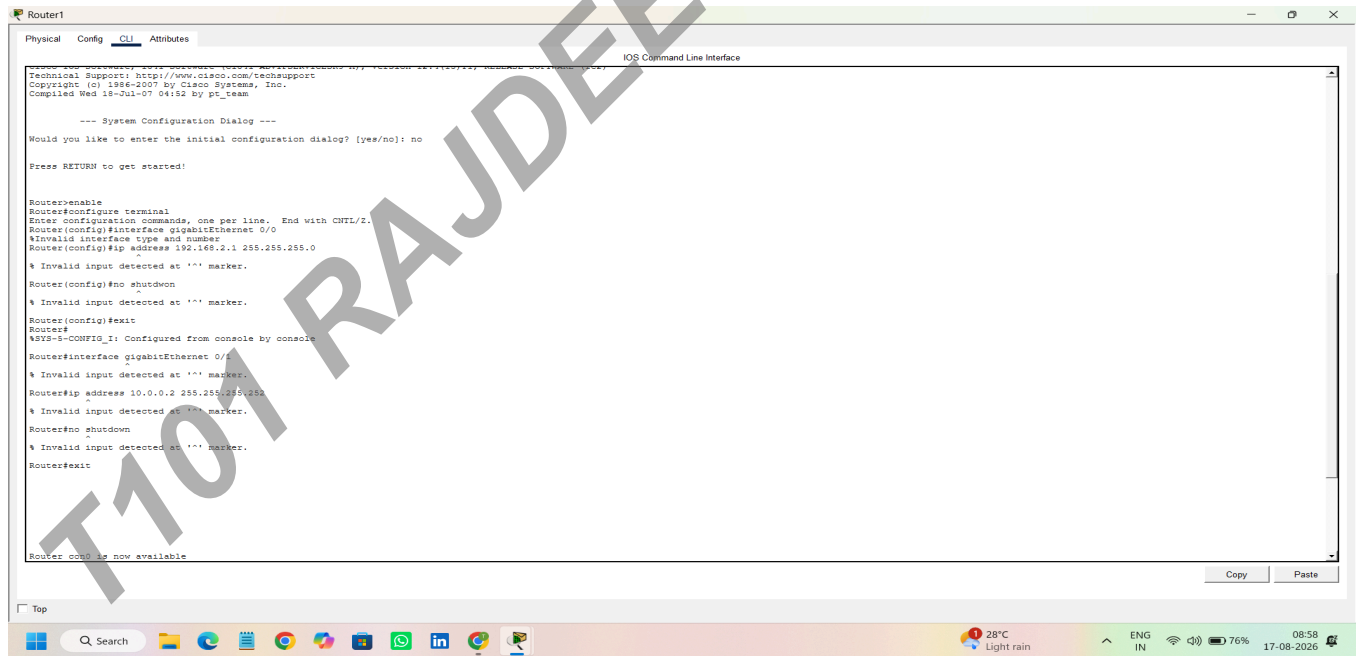
Press RETURN to get started!

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
%Invalid interface type and number
Router(config)#ip address 192.168.1.1 255.255.255.0
% Invalid input detected at '^' marker.
Router(config)#no shutdown
% Invalid input detected at '^' marker.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#ip address 10.0.0.1 255.255.255.252
% Invalid input detected at '^' marker.
Router#no shutdown
% Invalid input detected at '^' marker.
Router#exit
Router#exit

Router con0 is now available
```

Router 1:-



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
%Invalid interface type and number
Router(config)#ip address 192.168.2.1 255.255.255.0
% Invalid input detected at '^' marker.
Router(config)#no shutdown
% Invalid input detected at '^' marker.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#ip address 10.0.0.2 255.255.255.252
% Invalid input detected at '^' marker.
Router#no shutdown
% Invalid input detected at '^' marker.
Router#exit
Router#exit

Router con0 is now available
```

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The screenshot shows a Windows 10 desktop environment. A window titled 'PC0' is open, displaying the 'Cisco Packet Tracer PC Command Line 1.0' interface. The command prompt shows the following output:

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```

The taskbar at the bottom of the screen includes the Start button, a search bar, and several pinned application icons. The system tray on the right shows the date and time as 09:09 on 17-08-2026, along with icons for network, battery (72%), and language (ENG IN).

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

>
% Invalid input detected at '^' marker.
Router#no shutdown
% Invalid input detected at '^' marker.
Router#exit

Router con0 is now available

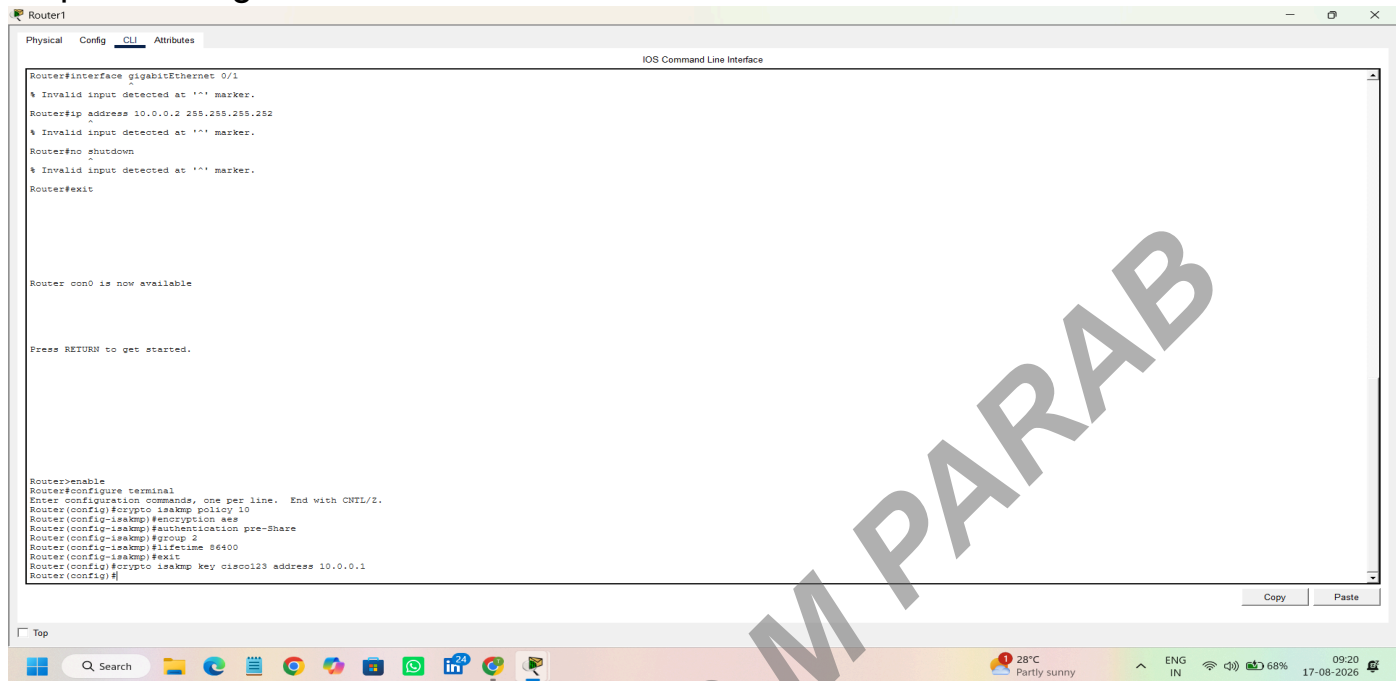
Press RETURN to get started.

Router>ip route 192.168.2.0 255.255.255.0 10.0.0.2
% Invalid input detected at '^' marker.
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#authentication aes
Router(config-isakmp)#sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.2
Router(config)#
```

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Step 6:- Configure IKE Phase 1 on Router1



The screenshot shows the CLI of Router1. The configuration includes setting the interface to gigabitEthernet 0/1, assigning IP address 10.0.0.2, and enabling the interface. It then enters the configuration mode for IKE Phase 1, setting the policy to 10, encryption to aes, authentication to pre-share, group to 2, lifetime to 86400, and the key to cisco123. The configuration is saved and the router is restarted. The status bar at the bottom shows the system is up and running.

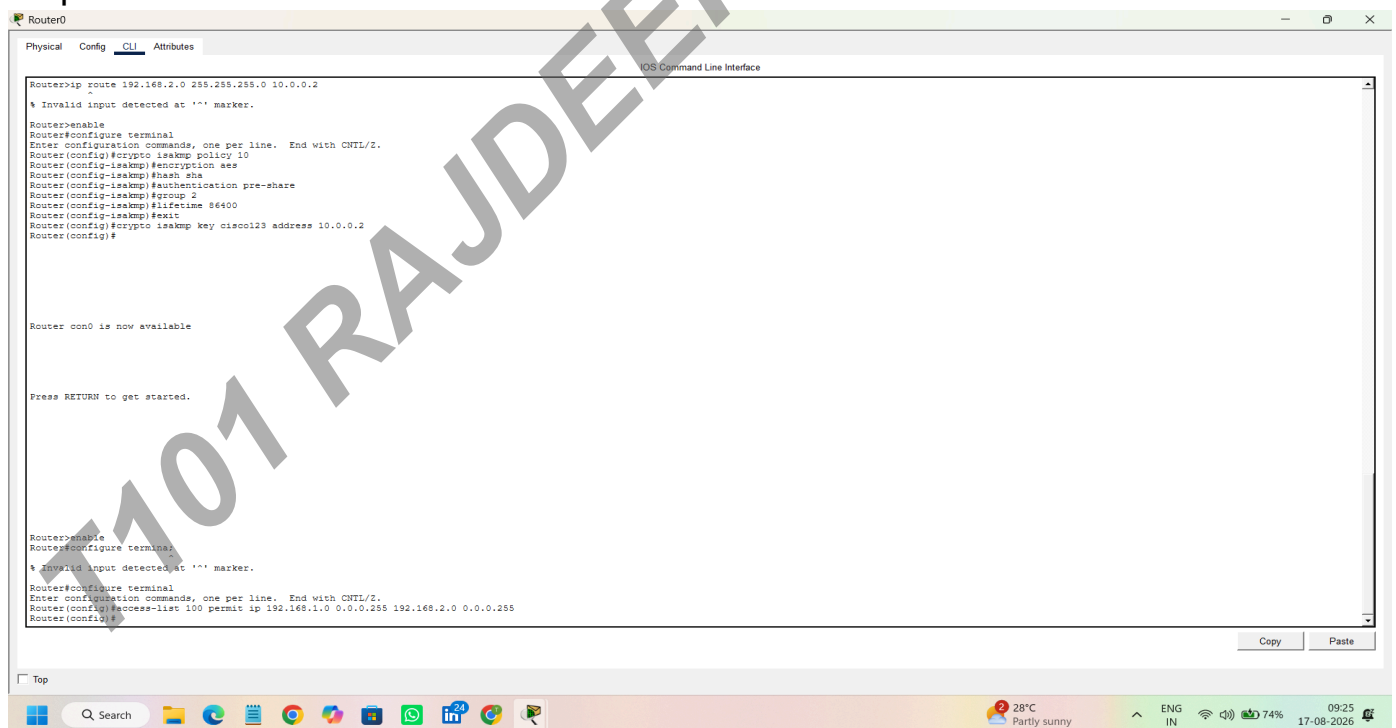
```
Router1>enable
Router1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router1(config)#ip address 10.0.0.2 255.255.255.252
% Invalid input detected at '^' marker.
Router1(config)#no shutdown
% Invalid input detected at '^' marker.
Router1(config)#exit

Router1>show ip interface brief
Router1 con0 is now available

Press RETURN to get started.

Router1>enable
Router1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#crypto isakmp policy 10
Router1(config-isakmp)#encryption aes
Router1(config-isakmp)#authentication pre-share
Router1(config-isakmp)#group 2
Router1(config-isakmp)#lifetime 86400
Router1(config-isakmp)#exit
Router1(config)#crypto isakmp key cisco123 address 10.0.0.1
Router1(config)#
```

Step 7:- Create the IPsec ACL on Router0



The screenshot shows the CLI of Router0. The configuration includes setting the interface to gigabitEthernet 0/0, assigning IP address 192.168.2.0, and enabling the interface. It then enters the configuration mode for IPsec ACL, setting the policy to 10, encryption to aes, authentication to pre-share, group to 2, lifetime to 86400, and the key to cisco123. The configuration is saved and the router is restarted. The status bar at the bottom shows the system is up and running.

```
Router0>enable
Router0#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router0(config)#interface gigabitEthernet 0/0
% Invalid input detected at '^' marker.
Router0(config)#ip address 192.168.2.0 10.0.0.2
% Invalid input detected at '^' marker.
Router0(config)#no shutdown
% Invalid input detected at '^' marker.
Router0(config)#exit

Router0>show ip interface brief
Router0 con0 is now available

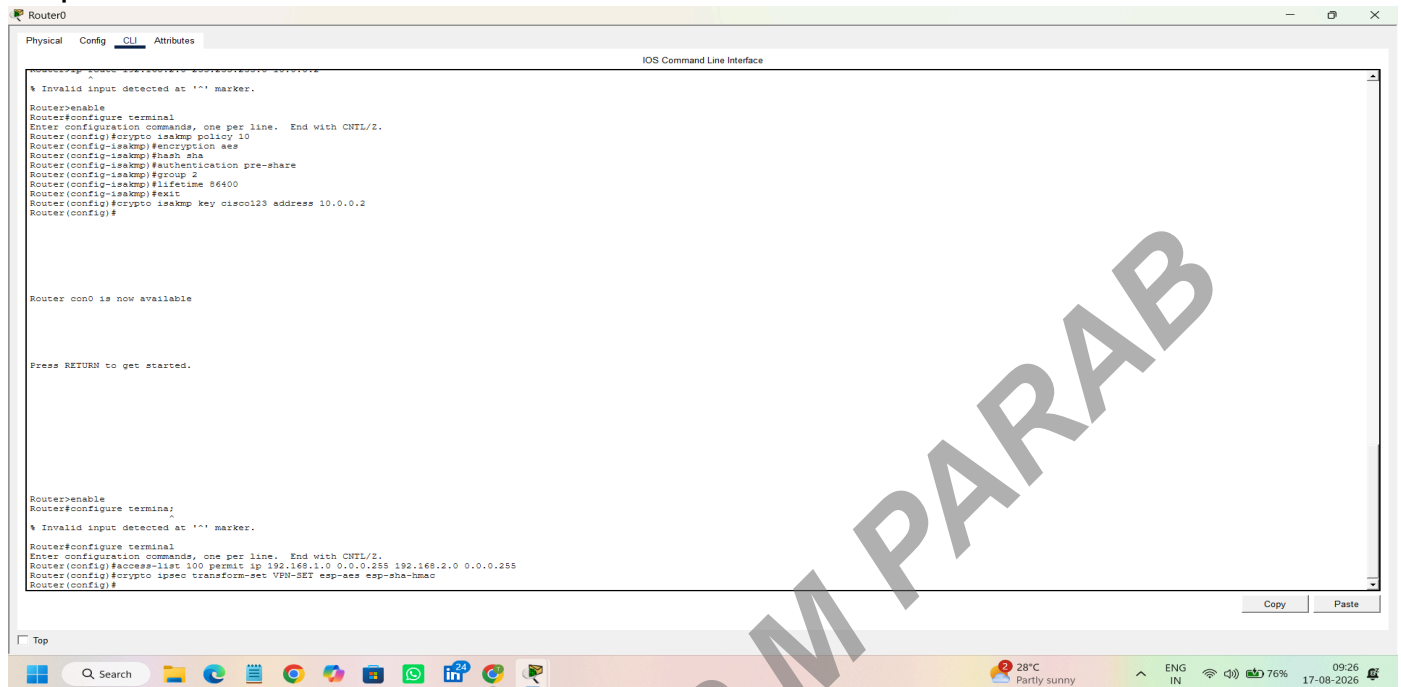
Press RETURN to get started.

Router0>enable
Router0#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router0(config)#crypto isakmp policy 10
Router0(config-isakmp)#encryption aes
Router0(config-isakmp)#authentication pre-share
Router0(config-isakmp)#group 2
Router0(config-isakmp)#lifetime 86400
Router0(config-isakmp)#exit
Router0(config)#crypto isakmp key cisco123 address 10.0.0.2
Router0(config)#
```

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Step 8:- Create the IPsec transform set on Router0



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router>
% Invalid input detected at '^' marker.

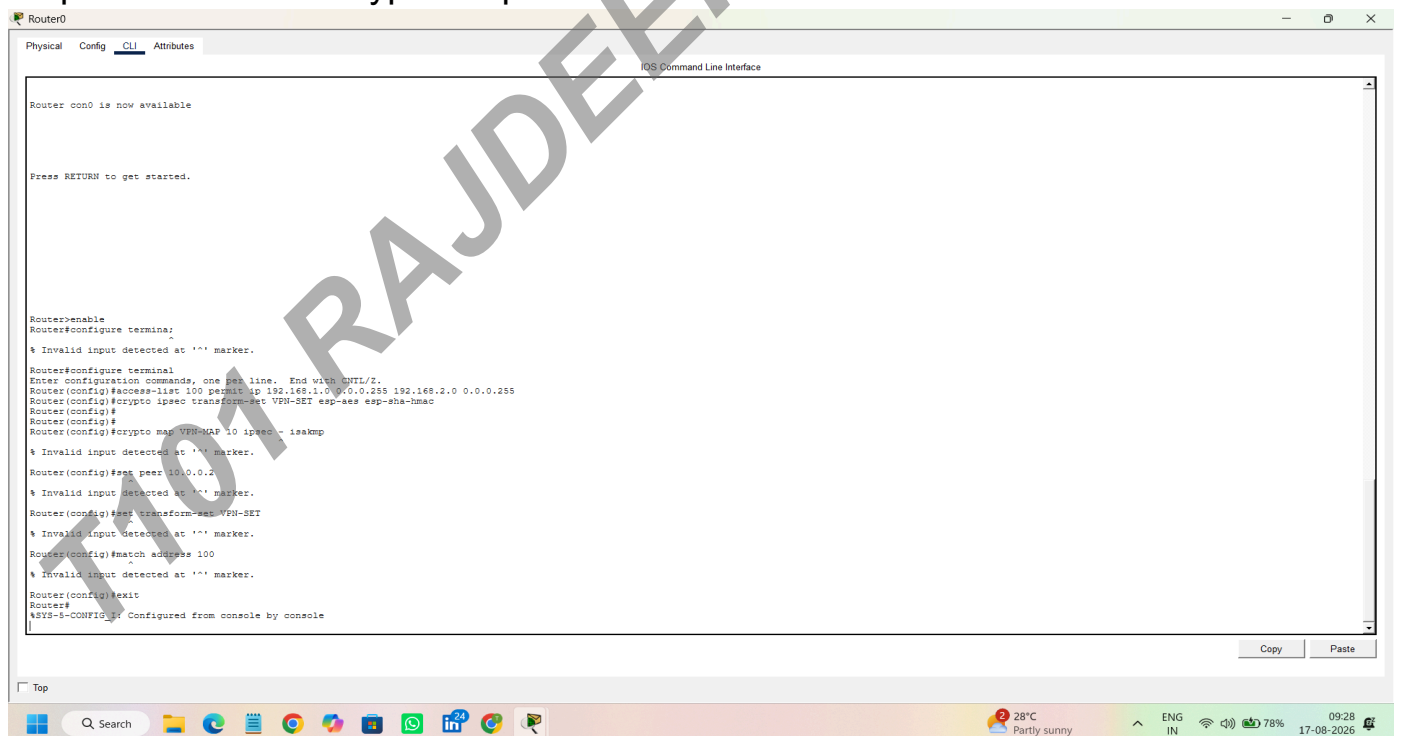
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.2
Router(config)#

Router con0 is now available

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#
```

Step 9:- Create the Crypto Map on Router0



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router con0 is now available

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
Router(config)#
% Invalid input detected at '^' marker.

Router(config)#set peer 10.0.0.2
% Invalid input detected at '^' marker.

Router(config)#set transform-set VPN-SET
% Invalid input detected at '^' marker.

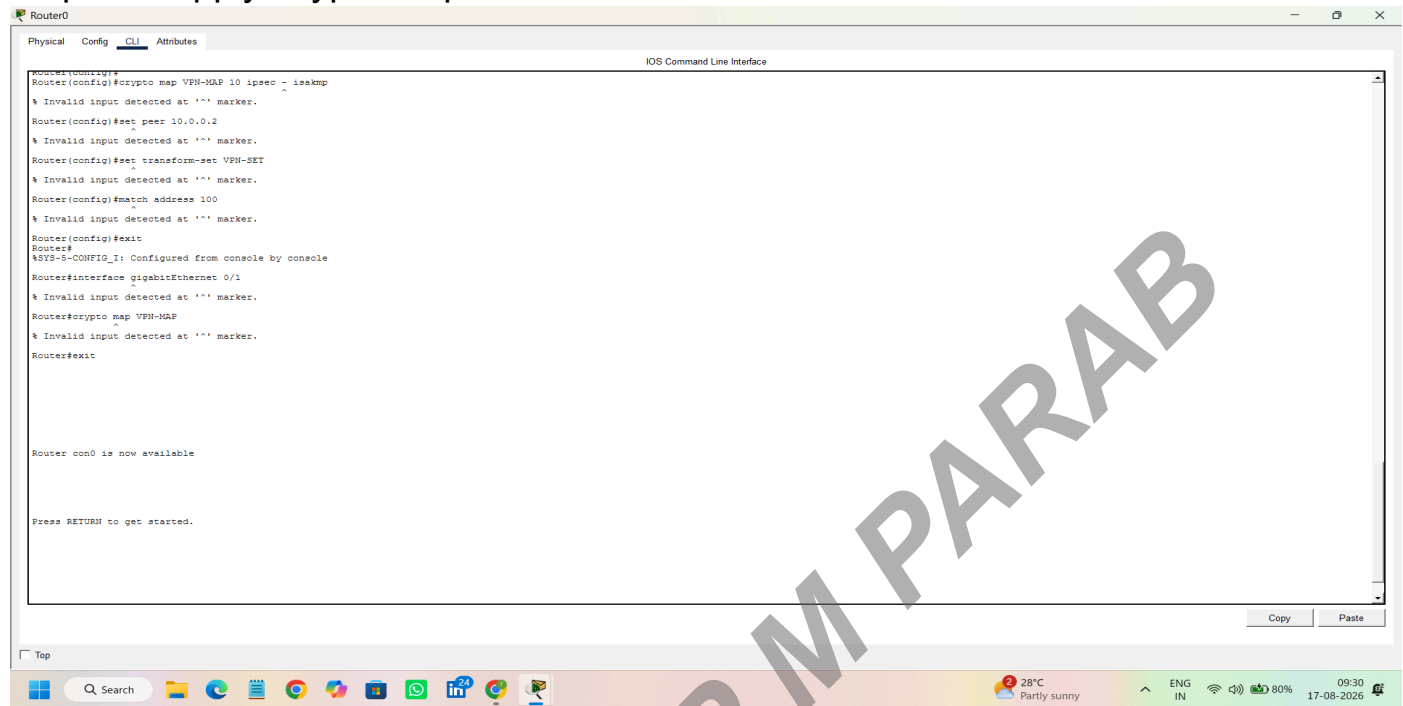
Router(config)#match address 100
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
SYS-5-CONFIG-2 Configured from console by console
```

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Step 10:- Apply Crypto Map to Router0 WAN interface

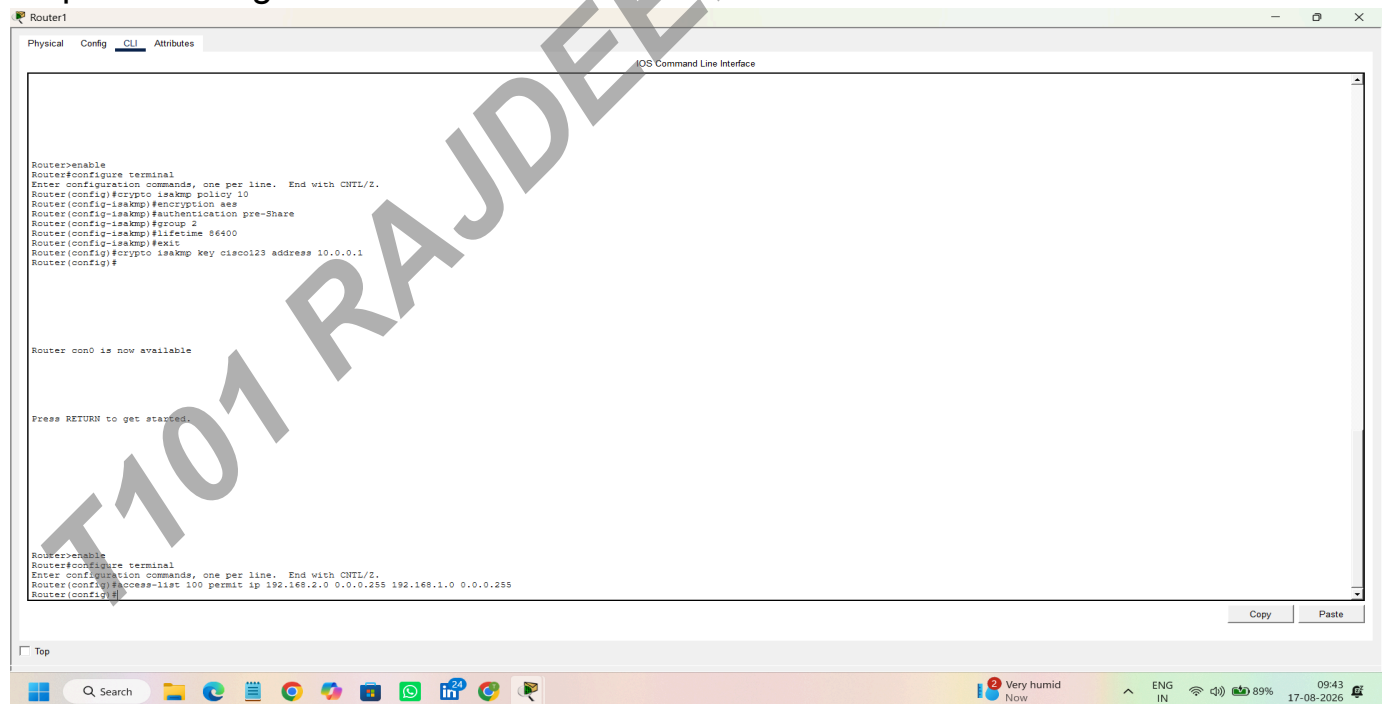
A screenshot of the Router0 CLI interface. The window title is "Router0". The tabs are "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, showing the "IOS Command Line Interface". The command history shows the following commands: Router(config)#crypto map VPN-MAP 10 ipsec - isakmp, Router(config)#set peer 10.0.0.2, Router(config)#set transform-set VPN-SET, Router(config)#match address 100, Router(config)#exit, Router#show sys-5-CONFIG_1: Configured from console by console, Router#interface gigabitEthernet 0/1, Router#crypto map VPN-MAP, and Router#exit. The output shows "Router con0 is now available" and "Press RETURN to get started." at the bottom. A large watermark "T101 RAJDEEP M PARAB" is visible across the screen.

```
Router(config)#crypto map VPN-MAP 10 ipsec - isakmp
% Invalid input detected at '^' marker.
Router(config)#set peer 10.0.0.2
% Invalid input detected at '^' marker.
Router(config)#set transform-set VPN-SET
% Invalid input detected at '^' marker.
Router(config)#match address 100
% Invalid input detected at '^' marker.
Router(config)#exit
Router#show sys-5-CONFIG_1: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#crypto map VPN-MAP
% Invalid input detected at '^' marker.
Router#exit

Router con0 is now available

Press RETURN to get started.
```

Step 11:- Configure IPsec ACL on Router1

A screenshot of the Router1 CLI interface. The window title is "Router1". The tabs are "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, showing the "IOS Command Line Interface". The command history shows the following commands: Router#enable, Router#configure terminal, Router(config)#crypto isakmp policy 10, Router(config-isakmp)#encryption aes, Router(config-isakmp)#authentication pre-Share, Router(config-isakmp)#group 2, Router(config-isakmp)#lifetime 86400, Router(config-isakmp)#exit, Router(config)#crypto isakmp key cisco123 address 10.0.0.1, Router(config)#, Router con0 is now available, Press RETURN to get started., Router#enable, Router#configure terminal, Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255, and Router(config)#. The output shows "Router con0 is now available" and "Press RETURN to get started." at the bottom. A large watermark "T101 RAJDEEP M PARAB" is visible across the screen.

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#authentication pre-Share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#

Router con0 is now available

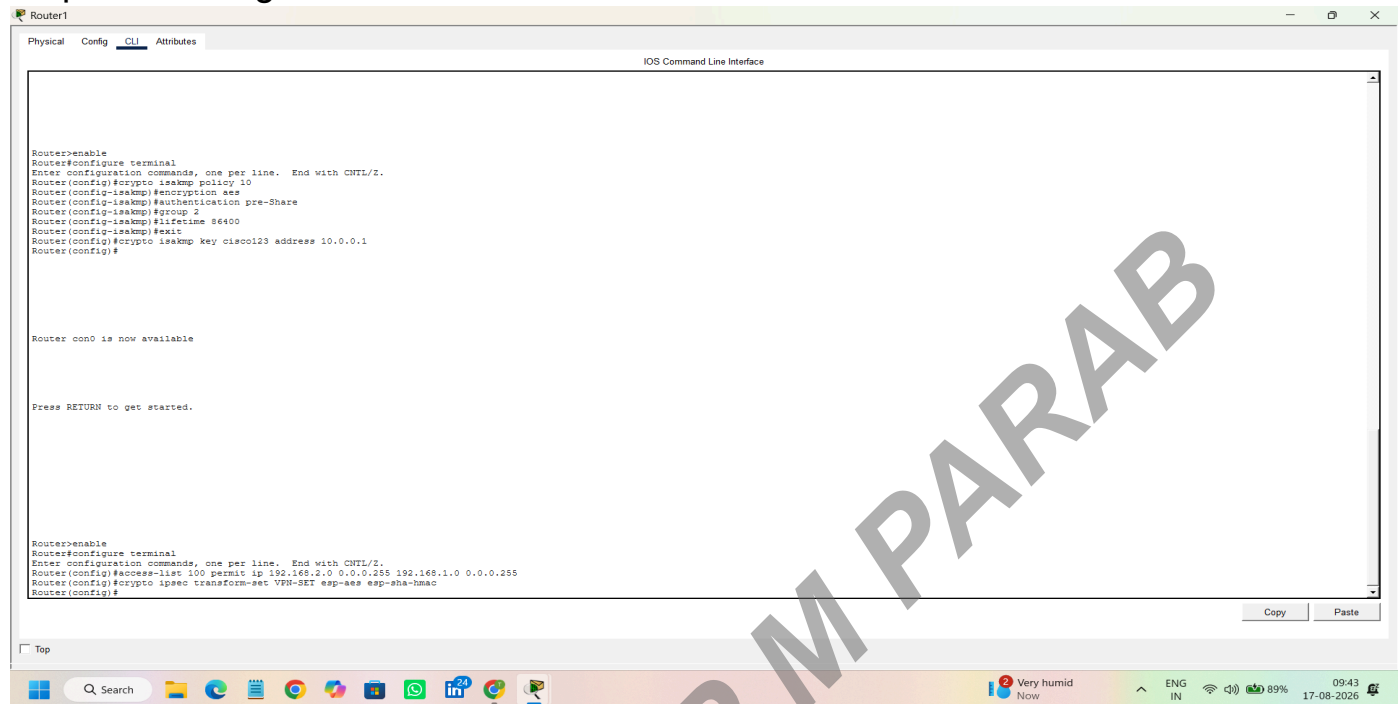
Press RETURN to get started.

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#
```

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Step 12:- Configure the Transform Set on Router1



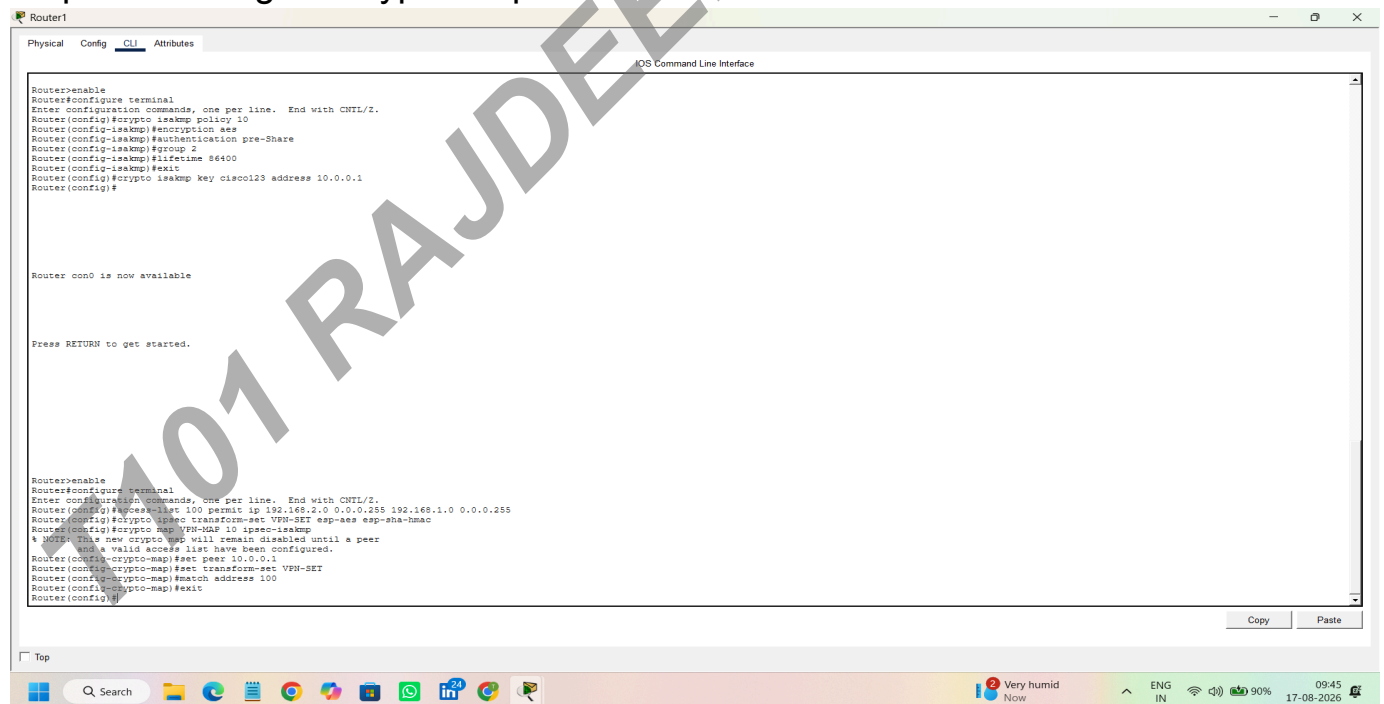
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#authentication pre-Share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#

Router con0 is now available.

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#
```

Step 13:- Configure Crypto Map on Router1



```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#authentication pre-Share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#

Router con0 is now available.

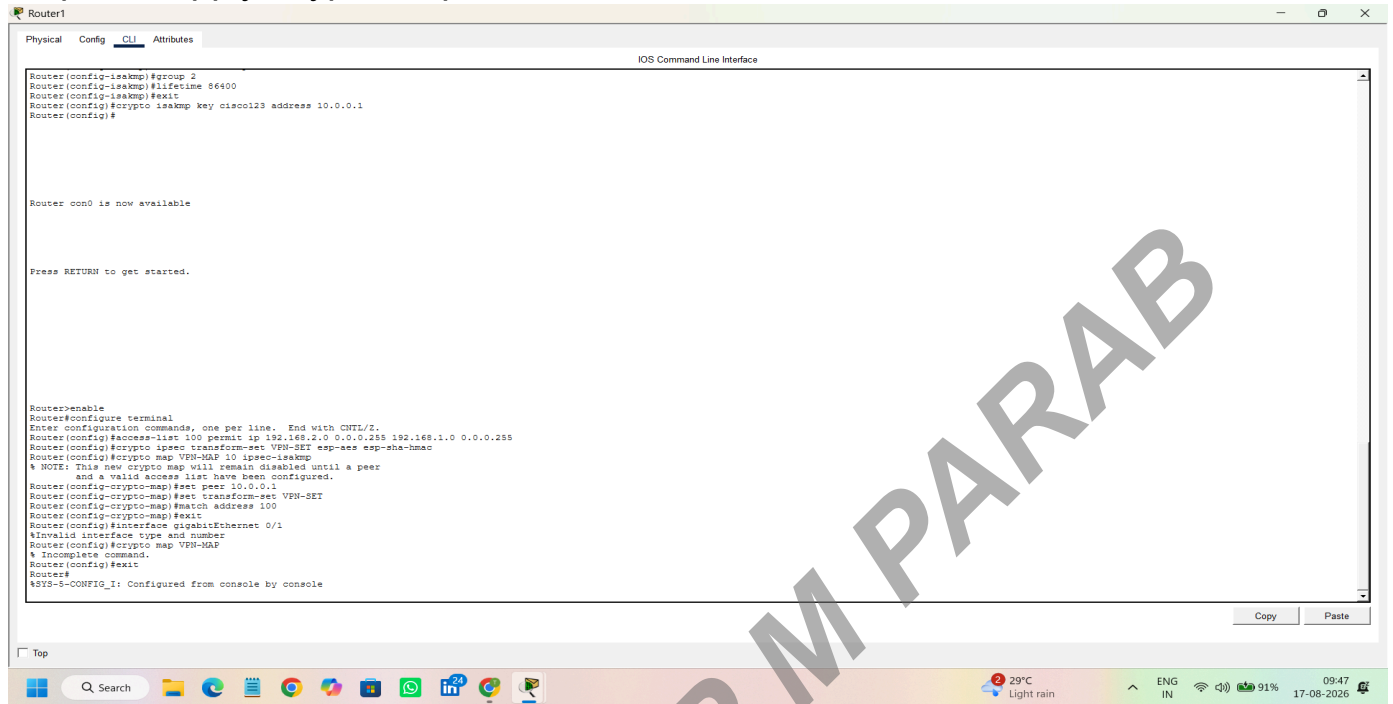
Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#crypto map VPN-IDS 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer with a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#
```


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Step 14:- Apply Crypto Map to Router1



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

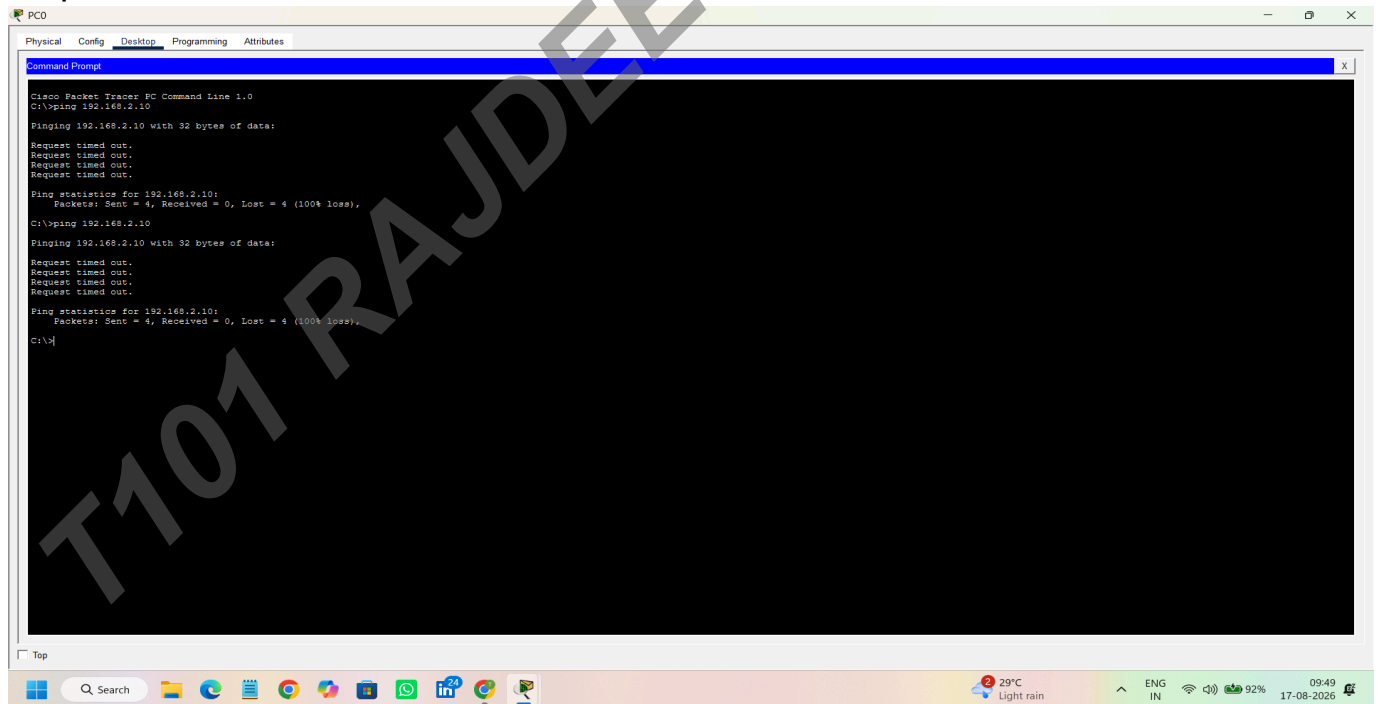
Router(config-isaamp)#group 2
Router(config-isaamp)#lifetime 86400
Router(config-isaamp)#exit
Router(config)#crypto isakmp key ciscot123 address 10.0.0.1
Router(config)#

Router con0 is now available.

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto isec transform-set VPN-SET esp-sec esp-sha-hmac
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
! NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#interface gigabitEthernet 0/1
%Invalid interface type and number
Router(config)#crypto map VPN-MAP
! Incomplete command.
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 15:- Generate traffic



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

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Step 16:- Verify IKE

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

% Invalid input detected at '^' marker.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#crypto map VPN-MAP
% Invalid input detected at '^' marker.
Router#exit

Router con0 is now available

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show crypto isakmp sa
% Invalid input detected at '^' marker.
Router(config)#
```

Step 17:- Verify IPsec

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

% Invalid input detected at '^' marker.
Router#crypto map VPN-MAP
% Invalid input detected at '^' marker.
Router#exit

Router con0 is now available

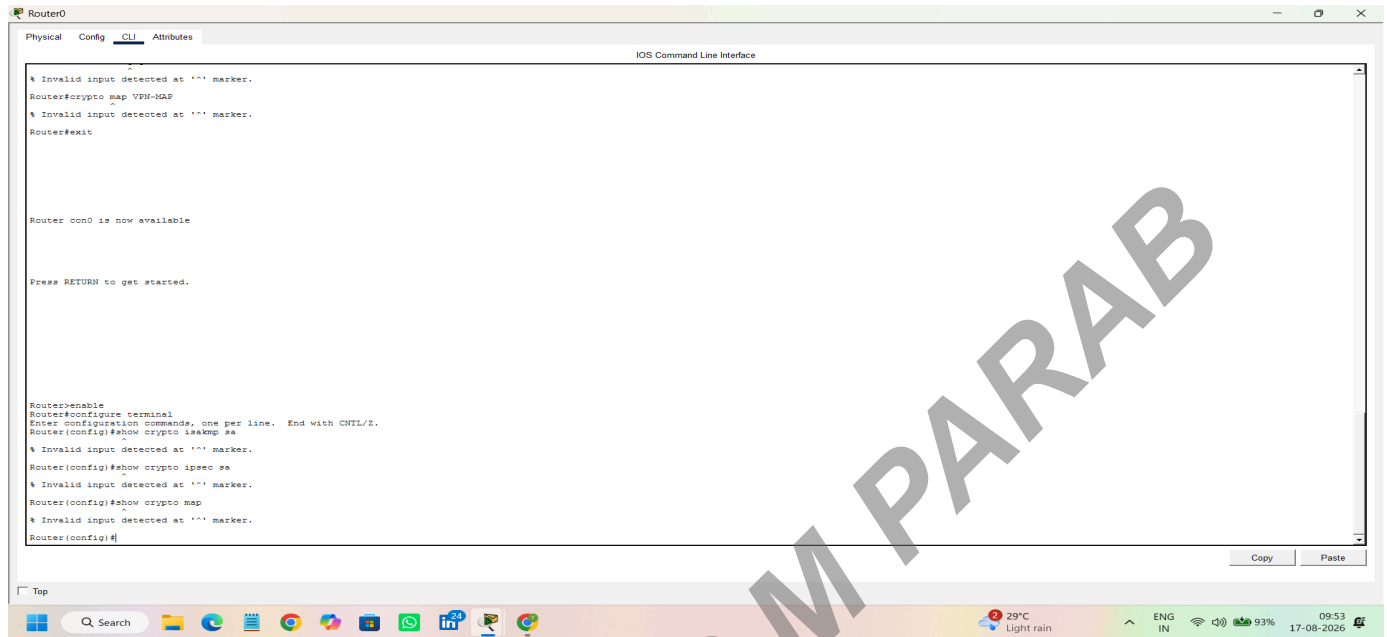
Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show crypto ipsec sa
% Invalid input detected at '^' marker.
Router(config)#show crypto ipsec
% Invalid input detected at '^' marker.
Router(config)#show crypto map
% Invalid input detected at '^' marker.
Router(config)#
```

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Step 18:- Verify the Crypto Map



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#
* Invalid input detected at '^' marker.
Router#crypto map VPN-MAP
* Invalid input detected at '^' marker.
Router#exit

Router#
Router con0 is now available

Press RETURN to get started.

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show crypto isakmp sa
* Invalid input detected at '^' marker.
Router(config)#show crypto ipsec sa
* Invalid input detected at '^' marker.
Router(config)#show crypto map
* Invalid input detected at '^' marker.
Router(config)#
```

The screenshot shows a Cisco Router CLI interface. The user is in the configuration mode (Router(config)#). They have entered the command 'show crypto isakmp sa' and 'show crypto ipsec sa', both of which resulted in an 'Invalid input detected' error. The user is currently at the 'Router(config)#' prompt.